

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z2,Z4

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z1 Z2 Z4: $\frac{Z_1Z_2Z_4g_m+Z_1Z_4}{2Z_1Z_2g_m+2Z_1+2Z_2+Z_4}$

$$H(z) = \frac{Z_1Z_2Z_4g_m + Z_1Z_4}{2Z_1Z_2g_m + 2Z_1 + 2Z_2 + Z_4}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1s}, \ R_2, \ \infty, \ \frac{R_4}{C_4R_4s+1}, \ \infty, \ \infty \right)$

$$H(s) = \frac{R_2R_4g_m + R_4}{2C_1C_4R_2R_4s^2 + 2R_2g_m + s(2C_1R_2 + C_1R_4 + 2C_4R_2R_4g_m + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_2}\sqrt{R_4}\sqrt{R_2g_m+1}}{2C_1R_2+C_1R_4+2C_4R_2R_4g_m+2C_4R_4}$
wo: $\frac{\sqrt{R_2g_m+1}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_2}\sqrt{R_4}}$
bandwidth: $\frac{2C_1R_2+C_1R_4+2C_4R_2R_4g_m+2C_4R_4}{2C_1C_4R_2R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \ R_2, \ \infty, \ \frac{1}{C_4s}, \ \infty, \ \infty \right)$

$$H(s) = \frac{R_1R_2g_m + R_1}{2C_1C_4R_1R_2s^2 + s(C_1R_1 + 2C_4R_1R_2g_m + 2C_4R_1 + 2C_4R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}}{C_1R_1+2C_4R_1R_2g_m+2C_4R_1+2C_4R_2}$
wo: $\frac{\sqrt{2}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_2g_m+2C_4R_1+2C_4R_2}{2C_1C_4R_1R_2}$
K-LP: $R_1R_2g_m + R_1$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_2 R_4 g_m + R_1 R_4}{2C_1 C_4 R_1 R_2 R_4 s^2 + 2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s(2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}\sqrt{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}}{2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}}$
 bandwidth: $\frac{2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4}{2C_1 C_4 R_1 R_2 R_4}$
 K-LP: $\frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_4 s + R_1 R_4 g_m}{2C_2 C_4 R_1 R_4 s^2 + 2R_1 g_m + s(2C_2 R_1 + C_2 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{2C_2 R_1 + C_2 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$
 bandwidth: $\frac{2C_2 R_1 + C_2 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4}{2C_2 C_4 R_1 R_4}$
 K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_4}{2C_2 R_1 + C_2 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{2C_2 C_4 R_1 R_2 s^2 + s(C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}}{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}$
 wo: $\frac{\sqrt{2}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}}$
 bandwidth: $\frac{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}{2C_2 C_4 R_1 R_2}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2}{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_4 s + R_1 R_2 R_4 g_m + R_1 R_4}{2C_2 C_4 R_1 R_2 R_4 s^2 + 2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s(2C_2 R_1 R_2 + C_2 R_2 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}\sqrt{2R_1R_2g_m+2R_1+2R_2+R_4}}{2C_2R_1R_2+C_2R_2R_4+2C_4R_1R_2R_4g_m+2C_4R_1R_4+2C_4R_2R_4} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1R_2g_m+2R_1+2R_2+R_4}}{2\sqrt{C_2}\sqrt{C_4}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2C_2R_1R_2+C_2R_2R_4+2C_4R_1R_2R_4g_m+2C_4R_1R_4+2C_4R_2R_4}{2C_2C_4R_1R_2R_4} \\ \text{K-LP: } & \frac{R_1R_2R_4g_m+R_1R_4}{2R_1R_2g_m+2R_1+2R_2+R_4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_1R_2R_4}{2C_2R_1R_2+C_2R_2R_4+2C_4R_1R_2R_4g_m+2C_4R_1R_4+2C_4R_2R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$8.4 \quad \mathbf{X-INVALID-NUMER-4} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2s}, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1R_4g_m + s(C_2R_1R_2R_4g_m + C_2R_1R_4)}{2R_1g_m + s^2(2C_2C_4R_1R_2R_4g_m + 2C_2C_4R_1R_4 + 2C_2C_4R_2R_4) + s(2C_2R_1R_2g_m + 2C_2R_1 + 2C_2R_2 + C_2R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_2}\sqrt{C_4}R_1R_2\sqrt{R_4g_m}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + 2\sqrt{C_2}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + 2\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}}}{2C_2R_1R_2g_m+2C_2R_1+2C_2R_2+C_2R_4+2C_4R_1R_4g_m+2C_4R_4} \\ \text{wo: } & \sqrt{\frac{R_1g_m+1}{C_2C_4R_1R_2R_4g_m+C_2C_4R_1R_4+C_2C_4R_2R_4}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_2C_4R_1R_2R_4g_m+C_2C_4R_1R_4+C_2C_4R_2R_4}}(2C_2R_1R_2g_m+2C_2R_1+2C_2R_2+C_2R_4+2C_4R_1R_4g_m+2C_4R_4)}{2\sqrt{C_2}\sqrt{C_4}R_1R_2\sqrt{R_4g_m}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + 2\sqrt{C_2}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}} + 2\sqrt{C_2}\sqrt{C_4}R_2\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_2g_m+R_1+R_2} + \frac{1}{R_1R_2g_m+R_1+R_2}}} \\ \text{K-LP: } & \frac{R_1R_4g_m}{2R_1g_m+2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_1R_2R_4g_m+C_2R_1R_4}{2C_2R_1R_2g_m+2C_2R_1+2C_2R_2+C_2R_4+2C_4R_1R_4g_m+2C_4R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$8.5 \quad \mathbf{X-INVALID-NUMER-5} \quad Z(s) = \left(\frac{1}{C_1s}, \quad \frac{1}{C_2s}, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2R_4s + R_4g_m}{C_1C_2R_4s^2 + 2g_m + s(2C_1 + 2C_2)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_4}\sqrt{g_m}}{2C_1+2C_2} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2C_1+2C_2}{C_1C_2R_4} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_4}{2C_1+2C_2} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$8.6 \quad \mathbf{X-INVALID-NUMER-6} \quad Z(s) = \left(\frac{1}{C_1s}, \quad \frac{1}{C_2s}, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2R_4s + R_4g_m}{2g_m + s^2(C_1C_2R_4 + 2C_1C_4R_4 + 2C_2C_4R_4) + s(2C_1 + 2C_2 + 2C_4R_4g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{R_4}\sqrt{g_m}\sqrt{C_1C_2+2C_1C_4+2C_2C_4}}{2C_1+2C_2+2C_4R_4g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_2R_4+2C_1C_4R_4+2C_2C_4R_4}} \\ \text{bandwidth: } & \frac{2C_1+2C_2+2C_4R_4g_m}{\sqrt{R_4}\sqrt{C_1C_2+2C_1C_4+2C_2C_4}\sqrt{C_1C_2R_4+2C_1C_4R_4+2C_2C_4R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_2R_4}{2C_1+2C_2+2C_4R_4g_m} \\ \text{Qz: } & \text{None} \end{aligned}$$

Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_2 R_4 s + R_2 R_4 g_m + R_4}{C_1 C_2 R_2 R_4 s^2 + 2 R_2 g_m + s(2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{R_4}\sqrt{R_2 g_m + 1}}{2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2}$
wo: $\frac{\sqrt{2} R_2 g_m + 2}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_2}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2 R_2 g_m + 2}(2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2)}{2 C_1 C_2 R_2 R_4 \sqrt{R_2 g_m + 1}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2 R_2 R_4}{2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2}$
Qz: None
Wz: None

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_2 R_4 s + R_2 R_4 g_m + R_4}{2 R_2 g_m + s^2 (C_1 C_2 R_2 R_4 + 2 C_1 C_4 R_2 R_4 + 2 C_2 C_4 R_2 R_4) + s(2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2 + 2 C_4 R_2 R_4 g_m + 2 C_4 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_2}\sqrt{R_4}\sqrt{R_2 g_m + 1}\sqrt{C_1 C_2 + 2 C_1 C_4 + 2 C_2 C_4}}{2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2 + 2 C_4 R_2 R_4 g_m + 2 C_4 R_4}$
wo: $\frac{\sqrt{2} R_2 g_m + 2}{\sqrt{C_1 C_2 R_2 R_4 + 2 C_1 C_4 R_2 R_4 + 2 C_2 C_4 R_2 R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2 R_2 g_m + 2}(2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2 + 2 C_4 R_2 R_4 g_m + 2 C_4 R_4)}{2 \sqrt{R_2}\sqrt{R_4}\sqrt{R_2 g_m + 1}\sqrt{C_1 C_2 + 2 C_1 C_4 + 2 C_2 C_4}\sqrt{C_1 C_2 R_2 R_4 + 2 C_1 C_4 R_2 R_4 + 2 C_2 C_4 R_2 R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2 R_2 R_4}{2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2 + 2 C_4 R_2 R_4 g_m + 2 C_4 R_4}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_4 g_m + s(C_2 R_2 R_4 g_m + C_2 R_4)}{2 g_m + s^2(2 C_1 C_2 R_2 + C_1 C_2 R_4) + s(2 C_1 + 2 C_2 R_2 g_m + 2 C_2)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{g_m}\sqrt{2 R_2 + R_4}}{2 C_1 + 2 C_2 R_2 g_m + 2 C_2}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2 C_1 C_2 R_2 + C_1 C_2 R_4}}$
bandwidth: $\frac{2 C_1 + 2 C_2 R_2 g_m + 2 C_2}{\sqrt{C_1}\sqrt{C_2}\sqrt{2 R_2 + R_4}\sqrt{2 C_1 C_2 R_2 + C_1 C_2 R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_2 R_2 R_4 g_m + C_2 R_4}{2 C_1 + 2 C_2 R_2 g_m + 2 C_2}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s(C_4 R_1 R_2 R_4 g_m + C_4 R_1 R_4)}{s^2(2 C_1 C_4 R_1 R_2 + C_1 C_4 R_1 R_4) + s(C_1 R_1 + 2 C_4 R_1 R_2 g_m + 2 C_4 R_1 + 2 C_4 R_2 + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{2R_2+R_4}}{C_1R_1+2C_4R_1R_2g_m+2C_4R_1+2C_4R_2+C_4R_4}$
 wo: $\frac{1}{\sqrt{2C_1C_4R_1R_2+2C_1C_4R_1R_4}}$
 bandwidth: $\frac{C_1R_1+2C_4R_1R_2g_m+2C_4R_1+2C_4R_2+C_4R_4}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{2R_2+R_4}\sqrt{2C_1C_4R_1R_2+C_1C_4R_1R_4}}$
 K-LP: $R_1R_2g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_4R_1R_2R_4g_m+C_4R_1R_4}{C_1R_1+2C_4R_1R_2g_m+2C_4R_1+2C_4R_2+C_4R_4}$
 Qz: None
 Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \frac{1}{C_2s}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_2R_1R_4s + R_1R_4g_m}{C_1C_2R_1R_4s^2 + 2R_1g_m + s(2C_1R_1 + 2C_2R_1 + C_2R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}}{2C_1R_1+2C_2R_1+C_2R_4}$
 wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_4}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_2R_1+C_2R_4)}{2C_1C_2R_1R_4\sqrt{R_1g_m+1}}$
 K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
 K-HP: 0
 K-BP: $\frac{C_2R_1R_4}{2C_1R_1+2C_2R_1+C_2R_4}$
 Qz: None
 Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \frac{1}{C_2s}, \infty, \frac{R_4}{C_4R_4s+1}, \infty, \infty \right)$

$$H(s) = \frac{C_2R_1R_4s + R_1R_4g_m}{2R_1g_m + s^2(C_1C_2R_1R_4 + 2C_1C_4R_1R_4 + 2C_2C_4R_1R_4) + s(2C_1R_1 + 2C_2R_1 + C_2R_4 + 2C_4R_1R_4g_m + 2C_4R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}\sqrt{C_1C_2+2C_1C_4+2C_2C_4}}{2C_1R_1+2C_2R_1+C_2R_4+2C_4R_1R_4g_m+2C_4R_4}$
 wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{C_1C_2R_1R_4+2C_1C_4R_1R_4+2C_2C_4R_1R_4}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_2R_1+C_2R_4+2C_4R_1R_4g_m+2C_4R_4)}{2\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}\sqrt{C_1C_2+2C_1C_4+2C_2C_4}\sqrt{C_1C_2R_1R_4+2C_1C_4R_1R_4+2C_2C_4R_1R_4}}$
 K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
 K-HP: 0
 K-BP: $\frac{C_2R_1R_4}{2C_1R_1+2C_2R_1+C_2R_4+2C_4R_1R_4g_m+2C_4R_4}$
 Qz: None
 Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \frac{R_2}{C_2R_2s+1}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_2R_1R_2R_4s + R_1R_2R_4g_m + R_1R_4}{C_1C_2R_1R_2R_4s^2 + 2R_1R_2g_m + 2R_1 + 2R_2 + R_4 + s(2C_1R_1R_2 + C_1R_1R_4 + 2C_2R_1R_2 + C_2R_2R_4)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}\sqrt{2R_1R_2g_m+2R_1+2R_2+R_4}}{2C_1R_1R_2+C_1R_1R_4+2C_2R_1R_2+C_2R_2R_4}$
 wo: $\frac{\sqrt{2R_1R_2g_m+2R_1+2R_2+R_4}}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_2}\sqrt{R_4}}$
 bandwidth: $\frac{2C_1R_1R_2+C_1R_1R_4+2C_2R_1R_2+C_2R_2R_4}{C_1C_2R_1R_2R_4}$
 K-LP: $\frac{R_1R_2R_4g_m+R_1R_4}{2R_1R_2g_m+2R_1+2R_2+R_4}$
 K-HP: 0
 K-BP: $\frac{C_2R_1R_2R_4}{2C_1R_1R_2+C_1R_1R_4+2C_2R_1R_2+C_2R_2R_4}$
 Qz: None
 Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{s^2 (C_1 C_2 R_1 R_2 + 2C_1 C_4 R_1 R_2 + 2C_2 C_4 R_1 R_2) + s (C_1 R_1 + C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + 2C_1 C_4 + 2C_2 C_4}}{C_1 R_1 + C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}$
 wo: $\frac{1}{\sqrt{C_1 C_2 R_1 R_2 + 2C_1 C_4 R_1 R_2 + 2C_2 C_4 R_1 R_2}}$
 bandwidth: $\frac{C_1 R_1 + C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}{\sqrt{R_1} \sqrt{R_2} \sqrt{C_1 C_2 + 2C_1 C_4 + 2C_2 C_4} \sqrt{C_1 C_2 R_1 R_2 + 2C_1 C_4 R_1 R_2 + 2C_2 C_4 R_1 R_2}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2}{C_1 R_1 + C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2}$
 Qz: None
 Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{C_2 R_1 R_2 R_4 s + R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s^2 (C_1 C_2 R_1 R_2 R_4 + 2C_1 C_4 R_1 R_2 R_4 + 2C_2 C_4 R_1 R_2 R_4) + s (2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_2 R_1 R_2 + C_2 R_2 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_2} \sqrt{R_4} \sqrt{C_1 C_2 + 2C_1 C_4 + 2C_2 C_4} \sqrt{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}}{2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_2 R_1 R_2 + C_2 R_2 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4}$
 wo: $\frac{\sqrt{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}}{\sqrt{C_1 C_2 R_1 R_2 R_4 + 2C_1 C_4 R_1 R_2 R_4 + 2C_2 C_4 R_1 R_2 R_4}}$
 bandwidth: $\frac{2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_2 R_1 R_2 + C_2 R_2 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4}{\sqrt{R_1} \sqrt{R_2} \sqrt{R_4} \sqrt{C_1 C_2 + 2C_1 C_4 + 2C_2 C_4} \sqrt{C_1 C_2 R_1 R_2 R_4 + 2C_1 C_4 R_1 R_2 R_4 + 2C_2 C_4 R_1 R_2 R_4}}$
 K-LP: $\frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2 R_4}{2C_1 R_1 R_2 + C_1 R_1 R_4 + 2C_2 R_1 R_2 + C_2 R_2 R_4 + 2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4}$
 Qz: None
 Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s (C_2 R_1 R_2 R_4 g_m + C_2 R_1 R_4)}{2R_1 g_m + s^2 (2C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_4) + s (2C_1 R_1 + 2C_2 R_1 R_2 g_m + 2C_2 R_1 + 2C_2 R_2 + C_2 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{2R_2 + R_4} \sqrt{R_1 g_m + 1}}{2C_1 R_1 + 2C_2 R_1 R_2 g_m + 2C_2 R_1 + 2C_2 R_2 + C_2 R_4}$
 wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{2C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_4}}$
 bandwidth: $\frac{\sqrt{2} \sqrt{2R_1 g_m + 2} (2C_1 R_1 + 2C_2 R_1 R_2 g_m + 2C_2 R_1 + 2C_2 R_2 + C_2 R_4)}{2\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{2R_2 + R_4} \sqrt{R_1 g_m + 1} \sqrt{2C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_4}}$
 K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 K-HP: 0
 K-BP: $\frac{C_2 R_1 R_2 R_4 g_m + C_2 R_1 R_4}{2C_1 R_1 + 2C_2 R_1 R_2 g_m + 2C_2 R_1 + 2C_2 R_2 + C_2 R_4}$
 Qz: None
 Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_2 R_4 g_m + R_4 + s (C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4)}{2R_2 g_m + s^2 (2C_1 C_4 R_1 R_2 R_4 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_2 R_4) + s (2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_4 R_2 R_4 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_1} \sqrt{C_4} R_1 R_2 \sqrt{R_4} g_m \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + 2\sqrt{C_1} \sqrt{C_4} R_1 \sqrt{R_4} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + 2\sqrt{C_1} \sqrt{C_4} R_2 \sqrt{R_4} \sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}}}{2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_4 R_2 R_4 g_m + 2C_4 R_4}$
 wo: $\sqrt{\frac{R_2 g_m + 1}{C_1 C_4 R_1 R_2 R_4 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_2 R_4}}$

bandwidth: $\frac{\sqrt{\frac{R_2 g_m + 1}{C_1 C_4 R_1 R_2 R_4 g_m + C_1 C_4 R_1 R_4 + C_1 C_4 R_2 R_4}} (2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_4 R_2 R_4 g_m + 2C_4 R_4)}{2\sqrt{C_1}\sqrt{C_4}R_1R_2\sqrt{R_4}g_m\sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}} + 2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{R_2 g_m}{R_1 R_2 g_m + R_1 + R_2} + \frac{1}{R_1 R_2 g_m + R_1 + R_2}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4}{2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_4 R_2 R_4 g_m + 2C_4 R_4}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, R_2, \infty, R_4, \infty, \infty)$

$$H(s) = \frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, R_2, \infty, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1}{s(2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2) + 1}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, R_2, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s(2C_4 R_1 R_2 R_4 g_m + 2C_4 R_1 R_4 + 2C_4 R_2 R_4)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, R_2, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{R_1 R_2 g_m + R_1 + s(C_4 R_1 R_2 R_4 g_m + C_4 R_1 R_4)}{s(2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4) + 1}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, R_4, \infty, \infty\right)$

$$H(s) = \frac{C_2 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s(2C_2 R_1 + C_2 R_4) + 2}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{2C_2 C_4 R_1 s^2 + s(C_2 + 2C_4 R_1 g_m + 2C_4)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \frac{1}{C_2 s}, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty\right)$

$$H(s) = \frac{C_2 C_4 R_1 R_4 s^2 + R_1 g_m + s(C_2 R_1 + C_4 R_1 R_4 g_m)}{s^2(2C_2 C_4 R_1 + C_2 C_4 R_4) + s(C_2 + 2C_4 R_1 g_m + 2C_4)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4, \infty, \infty\right)$

$$H(s) = \frac{C_2 R_1 R_2 R_4 s + R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s(2C_2 R_1 R_2 + C_2 R_2 R_4)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_2 R_1 R_2 R_4 g_m + C_2 R_1 R_4)}{2 R_1 g_m + s (2 C_2 R_1 R_2 g_m + 2 C_2 R_1 + 2 C_2 R_2 + C_2 R_4) + 2}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{s^2 (2 C_2 C_4 R_1 R_2 g_m + 2 C_2 C_4 R_1 + 2 C_2 C_4 R_2) + s (C_2 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_2 C_4 R_1 R_2 R_4 g_m + C_2 C_4 R_1 R_4) + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_4 R_1 R_4 g_m)}{s^2 (2 C_2 C_4 R_1 R_2 g_m + 2 C_2 C_4 R_1 + 2 C_2 C_4 R_2 + C_2 C_4 R_4) + s (C_2 + 2 C_4 R_1 g_m + 2 C_4)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 R_4 g_m + R_4}{2 R_2 g_m + s (2 C_1 R_2 + C_1 R_4) + 2}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 g_m + 1}{2 C_1 C_4 R_2 s^2 + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 g_m + s (C_4 R_2 R_4 g_m + C_4 R_4) + 1}{s^2 (2 C_1 C_4 R_2 + C_1 C_4 R_4) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 s + g_m}{2 C_4 g_m s + s^2 (C_1 C_2 + 2 C_1 C_4 + 2 C_2 C_4)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_4 s^2 + g_m + s (C_2 + C_4 R_4 g_m)}{C_1 C_2 C_4 R_4 s^3 + 2 C_4 g_m s + s^2 (C_1 C_2 + 2 C_1 C_4 + 2 C_2 C_4)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 R_2 s + R_2 g_m + 1}{s^2 (C_1 C_2 R_2 + 2 C_1 C_4 R_2 + 2 C_2 C_4 R_2) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_2 R_4 s^2 + R_2 g_m + s (C_2 R_2 + C_4 R_2 R_4 g_m + C_4 R_4) + 1}{C_1 C_2 C_4 R_2 R_4 s^3 + s^2 (C_1 C_2 R_2 + 2 C_1 C_4 R_2 + C_1 C_4 R_4 + 2 C_2 C_4 R_2) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m + s (C_2 R_2 g_m + C_2)}{2C_1 C_2 C_4 R_2 s^3 + 2C_4 g_m s + s^2 (C_1 C_2 + 2C_1 C_4 + 2C_2 C_4 R_2 g_m + 2C_2 C_4)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s (C_2 R_2 R_4 g_m + C_2 R_4)}{2C_1 C_2 C_4 R_2 R_4 s^3 + 2g_m s + s^2 (2C_1 C_2 R_2 + C_1 C_2 R_4 + 2C_1 C_4 R_4 + 2C_2 C_4 R_2 R_4 g_m + 2C_2 C_4 R_4) + s (2C_1 + 2C_2 R_2 g_m + 2C_2 + 2C_4 R_4 g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_2 C_4 R_2 R_4 g_m + C_2 C_4 R_4) + s (C_2 R_2 g_m + C_2 + C_4 R_4 g_m)}{2C_4 g_m s + s^3 (2C_1 C_2 C_4 R_2 + C_1 C_2 C_4 R_4) + s^2 (C_1 C_2 + 2C_1 C_4 + 2C_2 C_4 R_2 g_m + 2C_2 C_4)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4 + s (2C_1 R_1 R_2 + C_1 R_1 R_4)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{s^2 (C_1 C_2 R_1 + 2C_1 C_4 R_1 + 2C_2 C_4 R_1) + s (C_2 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_1 R_4 s^2 + R_1 g_m + s (C_2 R_1 + C_4 R_1 R_4 g_m)}{C_1 C_2 C_4 R_1 R_4 s^3 + s^2 (C_1 C_2 R_1 + 2C_1 C_4 R_1 + 2C_2 C_4 R_1 + C_2 C_4 R_4) + s (C_2 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_1 R_2 R_4 s^2 + R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 + C_4 R_1 R_2 R_4 g_m + C_4 R_1 R_4)}{C_1 C_2 C_4 R_1 R_2 R_4 s^3 + s^2 (C_1 C_2 R_1 R_2 + 2C_1 C_4 R_1 R_2 + C_1 C_4 R_1 R_4 + 2C_2 C_4 R_1 R_2 + C_2 C_4 R_2 R_4) + s (C_1 R_1 + C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{2C_1 C_2 C_4 R_1 R_2 s^3 + s^2 (C_1 C_2 R_1 + 2C_1 C_4 R_1 + 2C_2 C_4 R_1 R_2 g_m + 2C_2 C_4 R_1 + 2C_2 C_4 R_2) + s (C_2 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_2 R_1 R_2 R_4 g_m + C_2 R_1 R_4)}{2C_1 C_2 C_4 R_1 R_2 R_4 s^3 + 2R_1 g_m s + s^2 (2C_1 C_2 R_1 R_2 + C_1 C_2 R_1 R_4 + 2C_1 C_4 R_1 R_4 + 2C_2 C_4 R_1 R_2 R_4 g_m + 2C_2 C_4 R_1 R_4 + 2C_2 C_4 R_2 R_4) + s (2C_1 R_1 + 2C_2 R_1 R_2 g_m + 2C_2 R_1 + 2C_2 R_2 + C_2 R_4 + 2C_4 R_1 R_4 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_2 C_4 R_1 R_2 R_4 g_m + C_2 C_4 R_1 R_4) + s (C_2 R_1 R_2 g_m + C_2 R_1 + C_4 R_1 R_4 g_m)}{s^3 (2C_1 C_2 C_4 R_1 R_2 + C_1 C_2 C_4 R_1 R_4) + s^2 (C_1 C_2 R_1 + 2C_1 C_4 R_1 + 2C_2 C_4 R_1 R_2 g_m + 2C_2 C_4 R_1 + 2C_2 C_4 R_2 + C_2 C_4 R_4) + s (C_2 + 2C_4 R_1 g_m + 2C_4)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 R_4 g_m + R_4 + s (C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4)}{2 R_2 g_m + s (2 C_1 R_1 R_2 g_m + 2 C_1 R_1 + 2 C_1 R_2 + C_1 R_4) + 2}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1) + 1}{s^2 (2 C_1 C_4 R_1 R_2 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_2) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_2 g_m + s^2 (C_1 C_4 R_1 R_2 R_4 g_m + C_1 C_4 R_1 R_4) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_4 R_2 R_4 g_m + C_4 R_4) + 1}{s^2 (2 C_1 C_4 R_1 R_2 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_2 + C_1 C_4 R_4) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 s^2 + g_m + s (C_1 R_1 g_m + C_2)}{2 C_1 C_2 C_4 R_1 s^3 + 2 C_4 g_m s + s^2 (C_1 C_2 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_2 C_4)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_2 R_4)}{2 C_1 C_2 C_4 R_1 R_4 s^3 + 2 g_m + s^2 (2 C_1 C_2 R_1 + C_1 C_2 R_4 + 2 C_1 C_4 R_1 R_4 g_m + 2 C_1 C_4 R_4 + 2 C_2 C_4 R_4) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_2 + 2 C_4 R_4 g_m)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{1}{C_2 s}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 C_4 R_1 R_4 s^3 + g_m + s^2 (C_1 C_2 R_1 + C_1 C_4 R_1 R_4 g_m + C_2 C_4 R_4) + s (C_1 R_1 g_m + C_2 + C_4 R_4 g_m)}{2 C_4 g_m s + s^3 (2 C_1 C_2 C_4 R_1 + C_1 C_2 C_4 R_4) + s^2 (C_1 C_2 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_2 C_4)}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_2 s^2 + R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{2 C_1 C_2 C_4 R_1 R_2 s^3 + s^2 (C_1 C_2 R_2 + 2 C_1 C_4 R_1 R_2 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_2 + 2 C_2 C_4 R_2) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 R_1 R_2 R_4 s^2 + R_2 R_4 g_m + R_4 + s (C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4 + C_2 R_2 R_4)}{2 C_1 C_2 C_4 R_1 R_2 R_4 s^3 + 2 R_2 g_m + s^2 (2 C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_4 + 2 C_1 C_4 R_1 R_2 R_4 g_m + 2 C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_2 R_4 + 2 C_2 C_4 R_2 R_4) + s (2 C_1 R_1 R_2 g_m + 2 C_1 R_1 + 2 C_1 R_2 + C_1 R_4 + 2 C_2 R_2 + 2 C_4 R_2 R_4 g_m + 2 C_4 R_4) + 2}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \frac{R_2}{C_2 R_2 s + 1}, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_2 C_4 R_1 R_2 R_4 s^3 + R_2 g_m + s^2 (C_1 C_2 R_1 R_2 + C_1 C_4 R_1 R_2 R_4 g_m + C_1 C_4 R_1 R_4 + C_2 C_4 R_2 R_4) + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2 + C_4 R_2 R_4 g_m + C_4 R_4) + 1}{s^3 (2 C_1 C_2 C_4 R_1 R_2 + C_1 C_2 C_4 R_2 R_4) + s^2 (C_1 C_2 R_2 + 2 C_1 C_4 R_1 R_2 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_2 + C_1 C_4 R_4 + 2 C_2 C_4 R_2) + s (C_1 + 2 C_4 R_2 g_m + 2 C_4)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad \frac{1}{C_4 s}, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{2 C_4 g_m s + s^3 (2 C_1 C_2 C_4 R_1 R_2 g_m + 2 C_1 C_2 C_4 R_1 + 2 C_1 C_2 C_4 R_2) + s^2 (C_1 C_2 + 2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_2 C_4 R_2 g_m + 2 C_2 C_4)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_2 R_1 R_2 R_4 g_m + C_1 C_2 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_2 R_2 R_4 g_m + C_2 R_4)}{2g_m + s^3 (2C_1 C_2 C_4 R_1 R_2 R_4 g_m + 2C_1 C_2 C_4 R_1 R_4 + 2C_1 C_2 C_4 R_2 R_4) + s^2 (2C_1 C_2 R_1 R_2 g_m + 2C_1 C_2 R_1 + 2C_1 C_2 R_2 + C_1 C_2 R_4 + 2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_2 C_4 R_2 R_4 g_m + 2C_2 C_4 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_2 R_2 g_m + 2C_2 + 2C_4 R_4 g_m)}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{g_m + s^3 (C_1 C_2 C_4 R_1 R_2 R_4 g_m + C_1 C_2 C_4 R_1 R_4) + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1 + C_1 C_4 R_1 R_4 g_m + C_2 C_4 R_2 R_4 g_m + C_2 C_4 R_4) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2 + C_4 R_4 g_m)}{2C_4 g_m s + s^3 (2C_1 C_2 C_4 R_1 R_2 g_m + 2C_1 C_2 C_4 R_1 + 2C_1 C_2 C_4 R_2 + C_1 C_2 C_4 R_4) + s^2 (C_1 C_2 + 2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_2 C_4 R_2 g_m + 2C_2 C_4)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$

$$H(s) = \frac{C_2 C_4 R_1 R_2 R_4 s^2 + R_1 R_2 g_m + R_1 + s (C_2 R_1 R_2 + C_4 R_1 R_2 R_4 g_m + C_4 R_1 R_4)}{s^2 (2C_2 C_4 R_1 R_2 + C_2 C_4 R_2 R_4) + s (C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_2} \sqrt{C_4} \sqrt{R_2} \sqrt{2R_1 + R_4}}{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4}$
 wo: $\frac{1}{\sqrt{2C_2 C_4 R_1 R_2 + C_2 C_4 R_2 R_4}}$
 bandwidth: $\frac{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4}{\sqrt{C_2} \sqrt{C_4} \sqrt{R_2} \sqrt{2R_1 + R_4} \sqrt{2C_2 C_4 R_1 R_2 + C_2 C_4 R_2 R_4}}$
 K-LP: $R_1 R_2 g_m + R_1$
 K-HP: $\frac{R_1 R_4}{2R_1 + R_4}$
 K-BP: $\frac{C_2 R_1 R_2 + C_4 R_1 R_2 R_4 g_m + C_4 R_1 R_4}{C_2 R_2 + 2C_4 R_1 R_2 g_m + 2C_4 R_1 + 2C_4 R_2 + C_4 R_4}$
 Qz: None
 Wz: $\frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_2} \sqrt{C_4} \sqrt{R_2} \sqrt{R_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m + C_2 R_4)}{2g_m + s^2 (2C_1 C_2 R_1 + C_1 C_2 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_2)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_2} \sqrt{g_m} \sqrt{2R_1 + R_4}}{2C_1 R_1 g_m + 2C_1 + 2C_2}$
 wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{2C_1 C_2 R_1 + C_1 C_2 R_4}}$
 bandwidth: $\frac{2C_1 R_1 g_m + 2C_1 + 2C_2}{\sqrt{C_1} \sqrt{C_2} \sqrt{2R_1 + R_4} \sqrt{2C_1 C_2 R_1 + C_1 C_2 R_4}}$
 K-LP: $\frac{R_4}{2}$
 K-HP: $\frac{R_1 R_4}{2R_1 + R_4}$
 K-BP: $\frac{C_1 R_1 R_4 g_m + C_2 R_4}{2C_1 R_1 g_m + 2C_1 + 2C_2}$
 Qz: None
 Wz: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_2 R_4 s^2 + R_2 R_4 g_m + R_4 + s (C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4 + C_2 R_2 R_4)}{2R_2 g_m + s^2 (2C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_4) + s (2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_2 R_2) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{2R_1 + R_4} \sqrt{R_2 g_m + 1}}{2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_2 R_2}$
 wo: $\frac{\sqrt{2} \sqrt{R_2 g_m + 1}}{\sqrt{2C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_4}}$
 bandwidth: $\frac{\sqrt{2} \sqrt{2R_2 g_m + 1} (2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_2 R_2)}{2\sqrt{C_1} \sqrt{C_2} \sqrt{R_2} \sqrt{2R_1 + R_4} \sqrt{R_2 g_m + 1} \sqrt{2C_1 C_2 R_1 R_2 + C_1 C_2 R_2 R_4}}$

$$\begin{aligned}
\text{K-LP: } & \frac{R_4}{2} \\
\text{K-HP: } & \frac{R_1 R_4}{2R_1 + R_4} \\
\text{K-BP: } & \frac{C_1 R_1 R_2 R_4 g_m + C_1 R_1 R_4 + C_2 R_2 R_4}{2C_1 R_1 R_2 g_m + 2C_1 R_1 + 2C_1 R_2 + C_1 R_4 + 2C_2 R_2} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \frac{\sqrt{R_2 g_m + 1}}{\sqrt{C_1} \sqrt{C_2} \sqrt{R_1} \sqrt{R_2}}
\end{aligned}$$

$$10.4 \quad \mathbf{X-INVALID-WZ-4} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad R_2 + \frac{1}{C_2 s}, \quad \infty, \quad R_4, \quad \infty, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_2 R_1 R_2 R_4 g_m + C_1 C_2 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_2 R_2 R_4 g_m + C_2 R_4)}{2g_m + s^2 (2C_1 C_2 R_1 R_2 g_m + 2C_1 C_2 R_1 + 2C_1 C_2 R_2 + C_1 C_2 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_2 R_2 g_m + 2C_2)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1R_2g_m\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_2\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_4\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}}{2C_1R_1g_m+2C_1+2C_2R_2g_m+2C_2}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{2C_1C_2R_1R_2g_m+2C_1C_2R_1+2C_1C_2R_2+C_1C_2R_4}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_2R_1R_2g_m+2C_1C_2R_1+2C_1C_2R_2+C_1C_2R_4}}(2C_1R_1g_m+2C_1+2C_2R_2g_m+2C_2)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1R_2g_m\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_1\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_2\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_2}R_4\sqrt{\frac{g_m}{2R_1R_2g_m+2R_1+2R_2+R_4}}}$$

$$\begin{aligned}
\text{K-LP: } & \frac{R_4}{2} \\
\text{K-HP: } & \frac{R_1 R_2 R_4 g_m + R_1 R_4}{2R_1 R_2 g_m + 2R_1 + 2R_2 + R_4} \\
\text{K-BP: } & \frac{C_1 R_1 R_4 g_m + C_2 R_2 R_4 g_m + C_2 R_4}{2C_1 R_1 g_m + 2C_1 + 2C_2 R_2 g_m + 2C_2} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \sqrt{\frac{g_m}{C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1}}
\end{aligned}$$

11 X-PolynomialError